United States Patent [19]

Sooch

[11] **Patent Number: 4,550,284**

[45] **Date of Patent: Oct. 29, 1985**

[54] **MOS CASCODE CURRENT MIRROR**

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[21] Appl. No.: **610,881**

[22] Filed: **May 16, 1984**

[51] **Int. Cl.**[4] **G05F 3/08**

[52] **U.S. Cl.** **323/315;** 330/288

[58] **Field of Search** 323/312, 315; 330/288; 307/304, 297

[56] **References Cited**

U.S. PATENT DOCUMENTS

3,852,679	12/1974	Schade, Jr.	330/257
3,887,879	6/1974	Radovsky	330/257
4,281,261	7/1981	Adam	307/270
4,297,646	10/1981	LoCascio et al.	330/288
4,327,321	4/1982	Suzuki et al.	323/315
4,412,186	10/1983	Nagano	330/288
4,471,292	9/1984	Schneck et al.	323/315
4,477,782	10/1984	Swanson	330/288

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Attorney, Agent, or Firm—Wendy W. Koba

[57] **ABSTRACT**

An MOS current mirror is disclosed which comprises only two circuit branches and requires only a single reference current. The input circuit branch includes at least four MOS transistors (**40, 42, 44, 46**) connected in series and the output circuit branch includes at least two MOS transistors (**48, 50**) interconnected with selected transistors of the input circuit branch. Mirroring of the input current (I_{REF}) is accomplished by providing a transistor (**46, 50**) in each circuit branch with identical operating characteristics (V_{DS}, V_{GS}). High output impedance is achieved in accordance with the present invention by adjusting the channel constant (Z/L) of another transistor (**42**) in the input circuit branch to be one-third the value of the channel constant associated with each of the remaining transistors.

5 Claims, 4 Drawing Figures

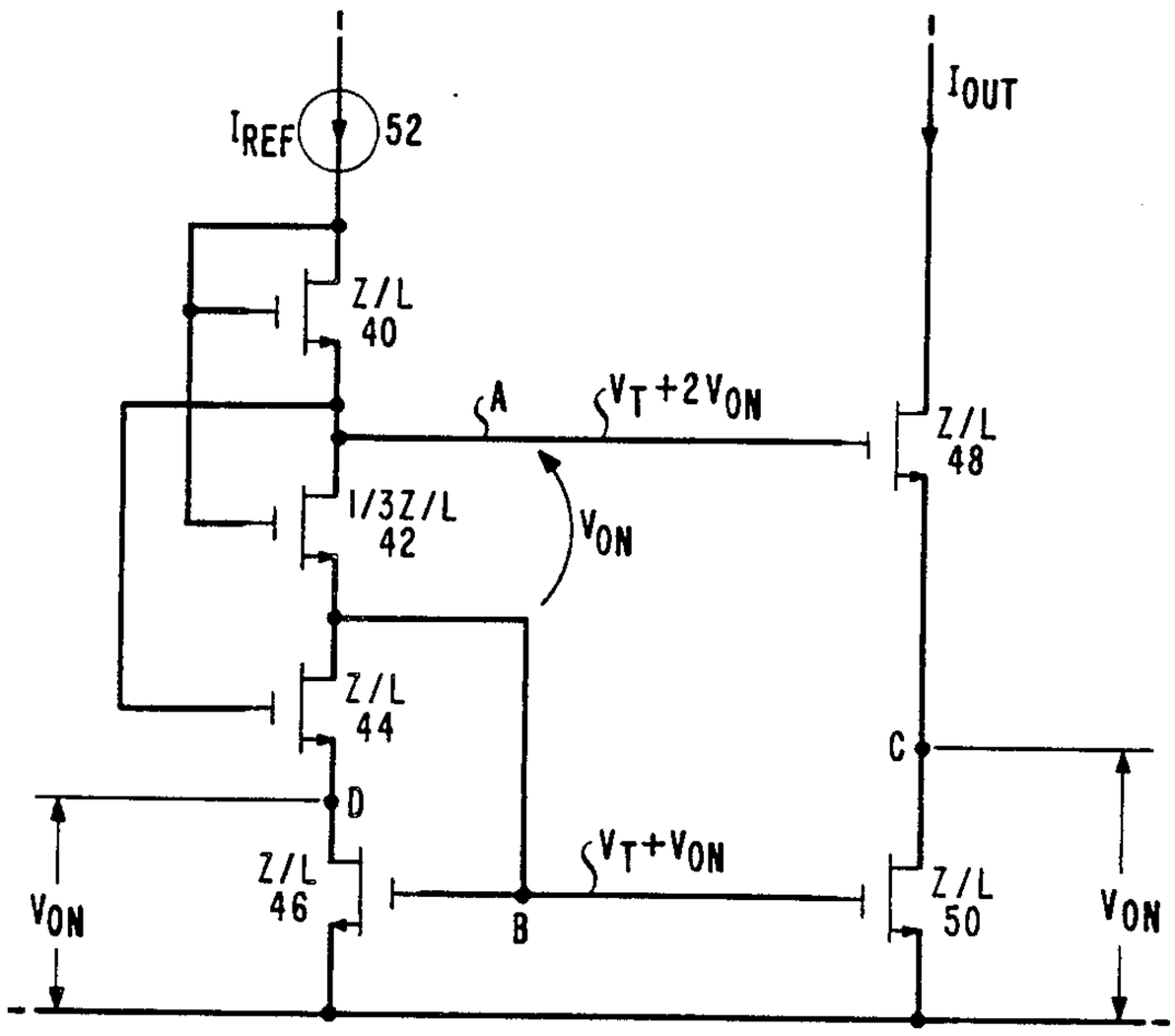


FIG. 1
(PRIOR ART)

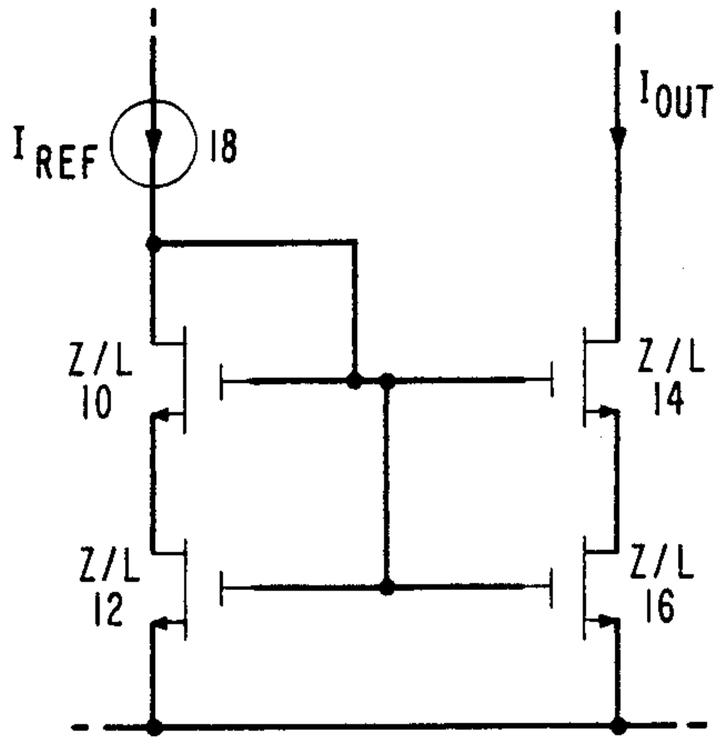


FIG. 2
(PRIOR ART)

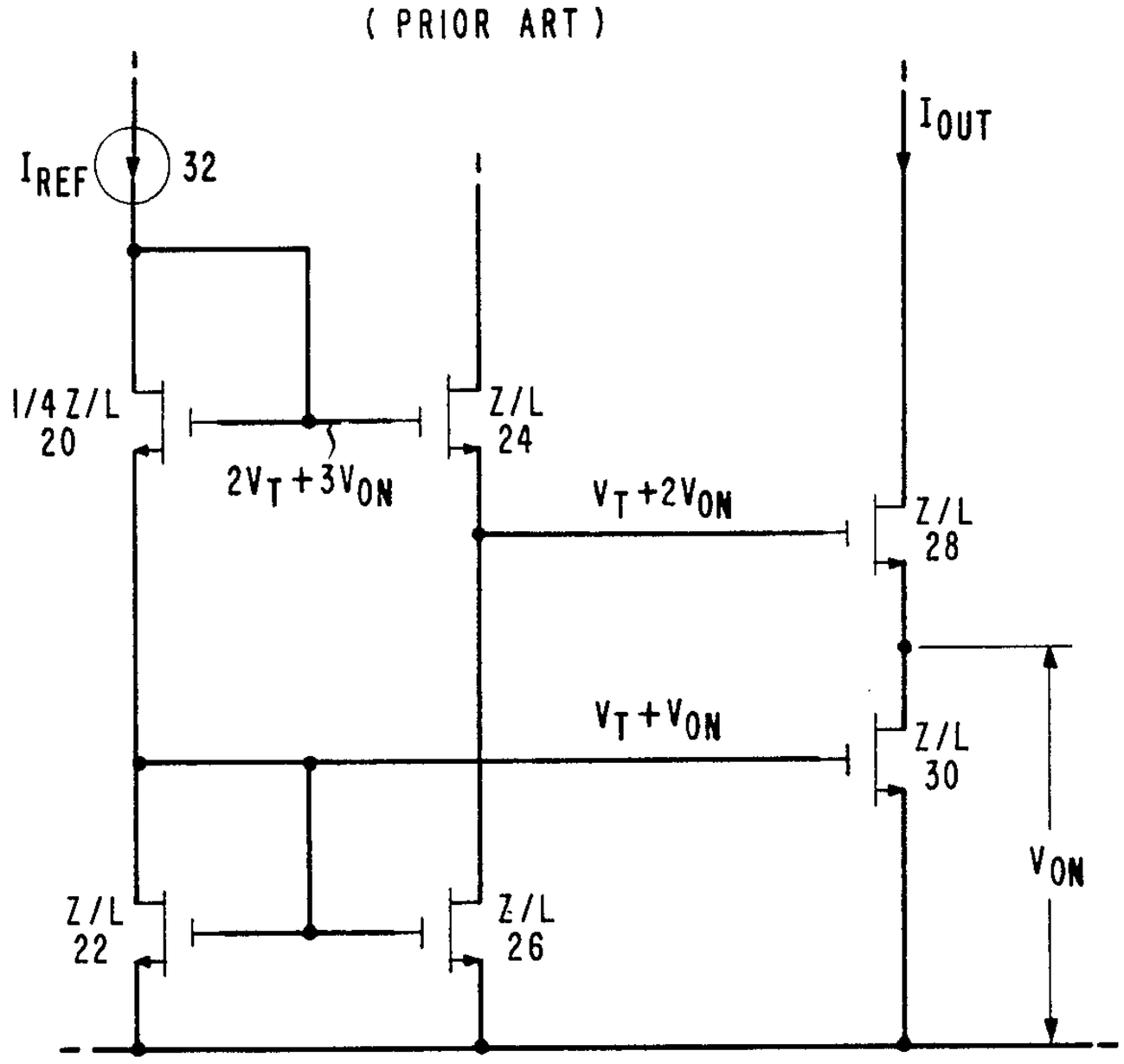


FIG. 3

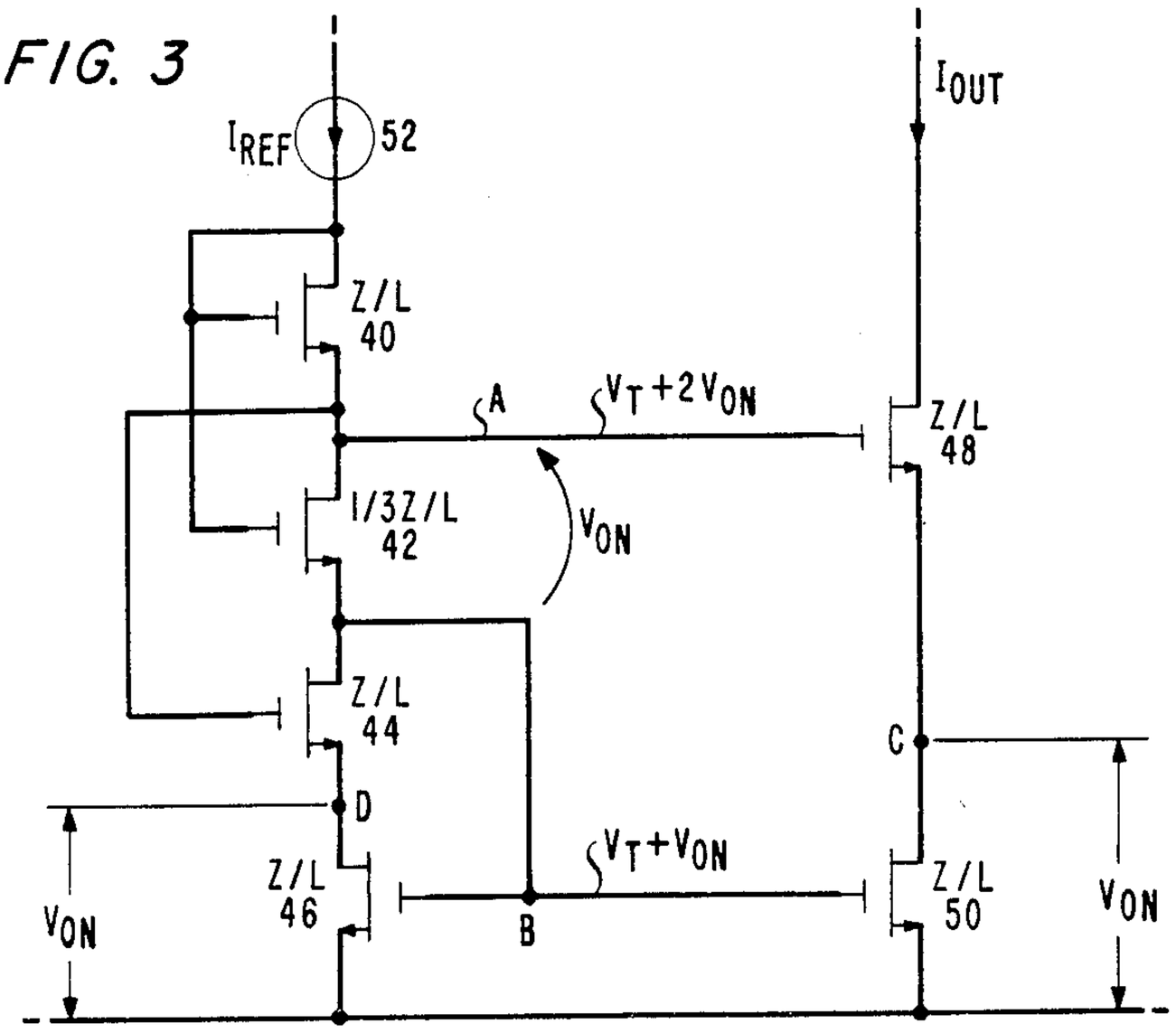
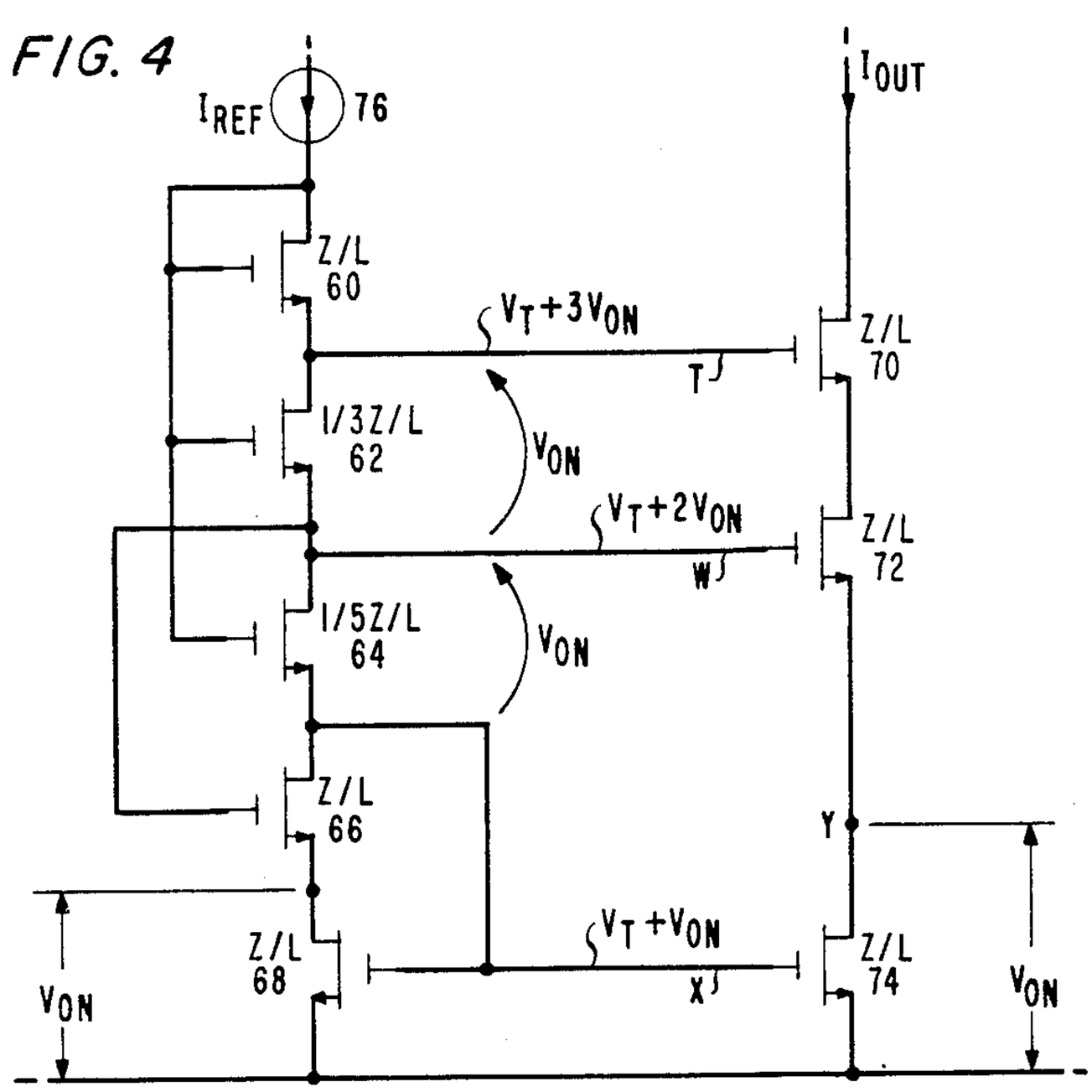

I_{REF}
52
Z/L
40
A
V_T+2V_{ON}
I_{OUT}
Z/L
48
1/3Z/L
42
V_{ON}
Z/L
44
C
D
V_T+V_{ON}
Z/L
46
Z/L
50
V_{ON}
B
V_{ON}

FIG. 4


I_{REF}
76
I_{OUT}
Z/L
60
V_T+3V_{ON}
T
Z/L
70
1/3Z/L
62
V_{ON}
V_T+2V_{ON}
W
Z/L
72
1/5Z/L
64
V_{ON}
Z/L
66
Y
V_T+V_{ON}
Z/L
68
Z/L
74
V_{ON}
X
V_{ON}

MOS CASCODE CURRENT MIRROR

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to an MOS current mirror and, more particularly, to an MOS cascode current mirror arrangement which requires only a single reference current while providing a large output impedance.

2. Description of the Prior Art

Current mirror circuits are well known in the art and have found uses in a variety of applications. Generally speaking, a current mirror circuit comprises a pair of transistors where an input reference current source is connected to drive one of the transistors. The pair of transistors are connected together in a manner whereby the reference current is substantially reproduced, or mirrored, at the output of the second transistor. In most cases, the critical factor in designing a current mirror circuit is providing optimum matching between the reference and output currents. U.S. Pat. No. 4,297,646 issued to LoCascio et al on Oct. 27, 1981 relates to a current mirror circuit, comprising bipolar transistors, with improved current matching provided by utilizing a single, split collector lateral bipolar transistor.

Current mirrors can also be formed using MOS devices, where one such arrangement is disclosed in U.S. Pat. No. 4,327,321 issued to H. Suzuki et al on Apr. 27, 1982. The Suzuki et al circuit also includes a resistor in the input rail between a P-channel MOSFET and an N-channel MOSFET to minimize the output current dependency on variations in the power supply. In MOS technology, small channel length devices are increasingly in demand. In relation to current mirror circuits, the decrease in channel length results in the decrease of the output impedance of the current mirror. Cascoding techniques become necessary, therefore, to increase the output impedance.

The advantages of cascoding transistors to form a stable current mirror are further exemplified in U.S. Pat. No. 4,412,186 issued to K. Nagano on Oct. 25, 1983. Like the LoCascio arrangement, Nagano discloses a current mirror circuit comprising bipolar transistors. In the Nagano arrangement, however, the circuit includes two stages, each having three transistors of one conductivity type and a fourth of the opposite conductivity type. When the four transistors are matched, the collector-to-emitter voltages, V_{CE}, of the third and fourth transistors are equivalent to their base-to-emitter voltages, V_{BE}.

These and other prior art cascode current mirror arrangements have not been widely used since they often exhibit one or more of the following problems; insufficient maximum voltage swing, excessive power consumption, insufficient output impedance, and inability to incorporate into integrated circuit designs.

SUMMARY OF THE INVENTION

The problems associated with prior art current mirrors are addressed by the present invention which relates to an MOS current mirror and, more particularly, to an MOS cascode current mirror arrangement which requires only a single reference current while providing a large output impedance.

It is an aspect of the present invention to provide current mirroring at a high output impedance with an arrangement of only six MOS transistors which requires only moderate power consumption, and can easily be incorporated into integrated circuits.

Another aspect of the present invention is to provide an MOS current mirror which can operate close to the circuit supply rails, thus providing a maximum output voltage swing.

A further aspect of this invention relates to maintaining an output impedance of at least g_m/g_o^2 while requiring only a single reference current source.

Other and further aspects of the present invention will become apparent during the course of the following discussion and by reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Referring now to the drawings:

FIG. **1** illustrates a basic prior art MOS cascode current mirror;

FIG. **2** illustrates an improved prior art MOS cascode current mirror which comprises three separate circuit branches;

FIG. **3** illustrates an MOS current mirror formed in accordance with the present invention; and

FIG. **4** illustrates an alternative MOS current mirror formed in accordance with the present invention.

DETAILED DESCRIPTION

A conventional prior art cascode current mirror, formed with MOS devices, is illustrated in FIG. **1**. As shown, an input circuit branch comprises an MOS transistor **10** connected in series with an MOS transistor **12** and an output circuit branch comprises an MOS transistor **14** connected in series with an MOS transistor **16**. The gates of transistors **10–16** are connected together as shown in FIG. **1**. A reference current **18**, denoted I_{REF}, is applied to the drain of transistor **10** and is subsequently reproduced, or mirrored, as the output current, I_{OUT}, at the drain of transistor **14**. Assuming that transistor **10–16** are well-matched, that is, they all have the same width-to-length channel ratio Z/L and are all connected to the same substrate, transistors **10** and **14** wil exhibit the same gate-to-source voltage, and similarly, transistors **12** and **16** will exhibit the same gate-to-source voltage. Therefore, since the current flowing through transistors **14** and **16** must match the current flowing through transistors **10** and **12**, I_{OUT} will be equal to, or mirror, the reference current I_{REF}. The current mirror illustrated in FIG. **1**, however, has a relatively low output impedance since transistor **16** will operate in its resistive region instead of its saturated region, thus lowering the impedance seen by the source of transistor **14**.

An alternative prior art arrangement, referred to as the Gray-Meyer cascode, which exhibits a relatively larger output impedance, is illustrated in FIG. **2**. As shown, an additional circuit branch is included in this arrangement. In the Gray-Meyer cascode, a pair of MOS transistors **20** and **22** form the input circuit branch and are connected in series where the gate of transistor **20** is connected to the drain of transistor **20** and similarly, the gate of transistor **22** is connected to the drain of transistor **22**. The next circuit branch contains a serially connected pair of MOS transistors **24** and **26**, where as shown in FIG. **2**, the gate of transistor **24** is connected to the gate of transistor **20** and the gate of transistor **26** is connected to the gate of transistor **22**. The remaining circuit branch, the output branch, includes a

pair of MOS transistors **28** and **30** also connected in series. In particular, the gate of transistor **28** is connected to the source of transistor **24** and the gate of transistor **30** is connected to the gates of transistors **22** and **26**. A reference current **32**, denoted I_{REF}, is applied to the drain of transistor **20** and is subsequently reproduced, or mirrored, at the drain of transistor **28**. To provide the higher output impedance, transistor **30** is biased on the edge of saturation, with its drain one threshold voltage, denoted V_T, more negative than its gate voltage, denoted V_T+V_{ON}, where V_{ON} is defined as the turn-on voltage of the device. This biasing is provided by transistors **24** and **26**, which generate the voltage V_T+2V_{ON} at the gate of transistor **28**. Transistor **20** is designed to comprise a channel width-to-length ratio one-fourth that of the remaining transistors to compensate for the addition of transistors **24** and **26**. The Gray-Meyer cascode does provide a high output impedance, but at the cost of a large power consumption, where the presence of the additional circuit branch is responsible for the increased power consumption. Further, the current I_{REF} will never be duplicated accurately in the middle circuit branch since the drain-to-source voltages of transistors **22** and **24** are inherently different.

An MOS cascode current mirror which exhibits a large output impedance and is formed in accordance with the present invention is illustrated in FIG. **3**. The arrangement shown, similar to the previous prior art circuits, includes N-channel MOS devices. However, it is to be understood that a current mirror formed in accordance with the present invention could also be formed from P-channel devices and the choice of N-channel devices in this instance is solely for the purpose of illustrating an exemplary embodiment of the invention.

As shown in FIG. **3**, a current mirror of the present invention comprises only two circuit branches, a first branch responsive to the input reference current and a second branch to replicate this current to provide the mirrored output current. In particular, the input branch comprises a series connection of four MOS transistors **40**, **42**, **44**, and **46**, and an input reference current **52**, denoted I_{REF}. As illustrated in FIG. **3**, the gate of transistor **40** is connected to its drain and also to the gate of transistor **42**. The gate of transistor **44** is connected to the source of transistor **40** and similarly, the gate of transistor **46** is connected to the source of transistor **42**. The output circuit branch of the present current mirror comprises a pair of serially connected MOS transistors **48** and **50**. As shown in FIG. **3**, the gate of transistor **48** is connected to the source of transistor **40** and the gate of transistor **44**, where this connection is defined as voltage node A, and the gate of transistor **50** is connected to the gate of transistor **46**, where this connection is defined as voltage node B.

In accordance with the present invention, reference current **52**, is coupled to the drain of transistor **40** and is subsequently reproduced as I_{OUT} along the output branch, as explained below in detail. It is to be noted that transistor **42** is formed to comprise a channel width-to-length ratio, Z/L, one-third that of the remaining transistors. The purpose of this size difference is critical to the performance of the present invention and will later be discussed in detail.

The basic premise of the present invention is to provide a current mirror with a large output impedance, where this results from creating a voltage at node A equal to V_T+2V_{ON} and a voltage at node B equal to V_T and V_{ON}. Following this premise, the voltage at node C, defined as the drain-to-source voltage (V_{DS}) of transistor **50**, will be equal to V_{ON}, since a V_T+V_{ON} voltage drop will occur between the gate and source of transistor **48**. Similarly, the voltage at node D, defined as V_{DS} of transistor **46**, will also be equal to V_{ON}, since a V_T+V_{ON} voltage drop will occur between the gate and source of transistor **44**. In accordance with the present invention, the gates of transistors **46** and **50** are connected together and are activated by the same gate to source voltage, V_{GS}, of V_T+V_{ON}. Since transistors **46** and **50**, as stated above, have the same V_{DS}, which is equal to V_{ON}, the same current will, be definition, flow through each device. Therefore, I_{OUT} will be identical to I_{REF}, that is, the output branch will mirror the current flowing through the input branch. Since the voltage at node A is forced to be V_T+2V_{ON}, the output circuit branch will exhibit a large output impedance.

Providing the required voltages at nodes A and B is accomplished using the process explained below. If all of the transistors illustrated in FIG. **3** are source-substrate connected, the threshold voltage, V_T, of each will be the same by definition. Providing a V_{DS} of transistor **43** equal to V_{ON} is accomplished by operating transistor **42** in its resistive region, where connecting the gates of transistors **40** and **42** forces transistor **42** to remain in its resistive region. Determining the necessary Z/L for transistor **42** is provided by the following calculations, where the current flowing through transistor **40** is assumed to be equal to the current flowing through transistor **42**, and V_{ON} is defined as the turn-on voltage of transistor **40**. Standard I-V relations for MOS devices results in

$$\frac{\mu}{2} C_o(Z/L)_1[2(V_{GS1} - V_T)V_{DS1} - V_{DS1}^2] = \frac{\mu}{2} C_o(Z/L)_2[V_{GS2} - V_T]^2, \quad (1)$$

where $(Z/L)_1$ is the channel constant of transistor **42**, V_{GS1} is the gate-to-source voltage of transistor **42**, V_{DS1} is the drain-to-source voltage of transistor **42**, $(Z/L)_2$ is the channel constant of transistor **40**, and V_{GS2} is the gate-to-source voltage of transistor **40**. If

$$V_{GS1} - V_T = 2V_{ON}, \text{ and} \quad (2)$$

$$V_{GS2} - V_T = V_{ON}, \quad (3)$$

as seen by reference to FIG. **3**, then

$$V_{DS1} = V_{GS1} - V_{GS2} = V_{ON}. \quad (4)$$

Substituting equations (2)–(4) into equation (1), and simplifying,

$$(Z/L)_1[2(2V_{ON})V_{ON} - V_{ON}^2] = (Z/L)_2[V_{ON}]^2. \quad (5)$$

Further simplification yields

$$(Z/L)_1[3V_{ON}^2] = (Z/L)_2[V_{ON}^2], \text{ or} \quad (6)$$

$$(Z/L)_1 = \tfrac{1}{3}(Z/L)_2. \quad (7)$$

Therefore, in accordance with equation (7) if all of the transistors are source-substrate connected and transistor **42** comprises a channel constant, Z/L, one-third

that of transistor **40**, the voltages necessary at nodes A and B to provide a high output impedance will be generated. If Z/L of transistor **42** is formed to be less than one-third the Z/L of transistor **40**, the voltage at node A will increase, thus insuring that transistor **50** operates well into its saturation region, providing an even greater output impedance. Additionally, if all of the transistors are not source-substrate connected, the Z/L of transistor **42** can be made as small as necessary to provide a V_{DS} of transistor **42** equal to V_{ON} and still provide a large output impedance. In general, the output impedance of this arrangement is defined by the quantity g_m/g_o^2, g_m is defined as the small signal transconductance and g_o is defined as the small signal output conductance. Additionally, the output voltage of this arrangement of the present invention can go as low as $2V_{ON}$ above the source of transistors **46** and **50** and still provide an output impedance of approximately g_m/g_o^2.

An even greater output impedance, on the order of g_m^2/g_o^3, at a minimum output voltage of $3V_{ON}$ can be obtained with an alternative circuit arrangement of the present invention, as illustrated in FIG. **4**. Similar to the previous embodiment, the current mirror illustrated in FIG. **4** comprises an input circuit branch and an output circuit branch. The input circuit branch includes a series connection of five MOS transistors **60–68**, and an input reference current source **76**, denoted I_{REF}. As seen by reference to FIG. **4**, the gate of transistor **60** is connected to its drain, and also to the gates of transistors **62** and **64**. The gate of transistor **66** is connected to the source of transistor **62** and similarly, the gate of transistor **68** is connected to the source of transistor **64**. The output circuit branch of the current mirror illustrated in FIG. **4** comprises a series connection of three MOS transistors **70–74**. As shown, the gate of transistor **70** is connected to the source of transistor **60**, where this connection is defined as voltage node T. Also, the gate of transistor **72** is connected to both the source of transistor **62** and the gate of transistor **66**, where this connection is defined as voltage node W. Lastly, at a voltage node X, the gate of transistor **74** is connected to both the gate of transistor **68** and the source of transistor **64**.

In accordance with the present invention, reference current **76**, is coupled to the drain of transistor **60** and is subsequently reproduced as I_{OUT} along the output circuit branch. It is to be noted that transistor **62** comprises a channel constant of $\frac{1}{3}$Z/L and transistor **64** comprises a channel constant of 1/5 Z/L in order to provide the voltages necessary at nodes T, W, and X to provide an output impedance of approximately g_m^2/g_o^3.

Applying the same premise as used in association with the description of the previous embodiment of the present invention, the voltage at node T must be equal to V_T+3V_{ON}, the voltage at node W equal to V_T+2V_{ON}, and the voltage at node X equal to V_T+V_{ON}. As before, current mirroring at a high output impedance will be achieved if transistors **68** and **74** comprise indentical characteristics. Here, the voltage at node Y, defined as V_{DS} of transistor **74**, will be equal to V_{ON}, since a V_T+V_{ON} voltage drop will occur between the gate and source of transistor **72**. V_{DS} of transistor **68** is also equal to V_{ON}, since a V_T+V_{ON} voltage drop will occur between the gate and source of transistor **66**. Since the gates of transistors **68** and **74** are coupled together and connected to the source of transistor **64**, and each has the same V_{DS}, which is equal to V_{ON}, the same current will, by definition, flow through transistors **68** and **74**, thus forcing I_{OUT} to be equal to I_{REF}.

In order to provide the necessary voltages at nodes T, W, and X, the same process as described above in association with FIG. **3** must be followed. Again, for the purposes of the present discussion it will be assumed that all of the devices are source-substrate connected so that each has the same threshold voltage V_T. Providing V_{DS} of transistors **62** and **64** to be equal to V_{ON} is accomplished by operating both transistors in their resistive region, as a result of connecting the gates of transistors **62** and **64** to the gate of transistor **60**. To determine the necessary Z/L for transistors **62** and **64**, the current flowing through transistors **64**, **62**, and **60**, defined as I_1, I_2, and I_3, respectively, are set equal to each other, where this can be expressed by the following relation

$$\mu\frac{C_o}{2}(Z/L)_1[2(V_{GS1}-V_T)V_{DS1}-V_{DS1}^2]= \tag{8}$$

$$\mu\frac{C_o}{2}(Z/L)_2[2(V_{GS2}-V_T)V_{DS2}-V_{DS2}^2]=$$

$$\mu\frac{C_o}{2}(Z/L)_3[V_{GS3}-V_T]^2.$$

If

$$V_{GS1}-V_T=3V_{ON}, \tag{9}$$

$$V_{GS2}-V_T=2V_{ON}, \text{ and} \tag{10}$$

$$V_{GS3}-V_T=V_{ON}, \tag{11}$$

then as seen by reference to FIG. **4**,

$$V_{DS1}=V_{GS1}-V_{GS2}=V_{ON}, \text{ and} \tag{12}$$

$$V_{DS2}=V_{GS2}-V_{GS3}=V_{ON}. \tag{13}$$

Substituting equations (9)–(13) into equation (8) and simplifying,

$$(Z/L)_1[2(3V_{ON})V_{ON}-V_{ON}^2]=(Z/L)_2[2(2V_{ON})-V_{ON}-V_{ON}^2]=(Z/L)_3[V_{ON}]^2. \tag{14}$$

Further simplification yields

$$(Z/L)_1[5V_{ON}^2]=(Z/L)_2[3V_{ON}^2]=(Z/L)_3[V_{ON}^2], \text{ or} \tag{15}$$

$$(Z/L)_1=1/5(Z/L)_3, \text{ and} \tag{16}$$

$$(Z/L)_2=\tfrac{1}{3}(Z/L)_3. \tag{17}$$

Therefore, in accordance with the present invention, if transistor **62** comprises a Z/L one-third that of transistor **60**, and transistor **64** comprises a Z/L one-fifth that of transistor **60**, the voltages V_T+3V_{ON}, V_T+2V_{ON}, and V_T+V_{ON} can be generated at nodes T, W, and X, respectively, thereby providing on MOS current mirror with an output impedance on the order of g_m^2/g_o^3.

What is claimed is:

1. An MOS current mirror including an input circuit branch and an output circuit branch, where the input circuit branch is responsive to a reference current and the output circuit branch mirrors the reference current to produce an output current substantially equal to said reference current, the input branch comprising

a series connection of a plurality of four MOS transistors each MOS transistor having a gate, a source,

and a drain electrode, where the drain of the first MOS transistor is responsive to said reference current and the gate of said first MOS transistor is connected to both said drain and the gate of the second MOS transistor, the gate of the third MOS transistor is connected to both the source of said first MOS transistor and the drain of said second MOS transistor, and the gate of the fourth MOS transistor is connected to both the source of said second MOS transistor and the drain of the third MOS transistor; and

the output branch comprising

a pair of MOS transistors each MOS transistor having a gate, a source, and a drain electrode, where the gate of a first MOS transistor of said pair of MOS transistors is connected to the source of said first MOS transistor of said input circuit branch and the gate of the remaining MOS transistor of said pair of MOS transistors is connected to the gate of said fourth MOS transistor of said input circuit branch, wherein

each MOS transistor of both said input and said output circuit branches comprises a channel constant defined by its associated channel width divided by its associated channel length, wherein the channel constant of said second MOS transistor of said input circuit branch is at most one-third the value of the channel constant associated with the remaining MOS transistors, where each of the remaining MOS transistors comprises substantially identical channel constant.

2. An MOS current mirror as defined in claim **1** wherein

the input circuit branch further comprises an additional MOS transistor connected in series between the second and the third MOS transistors, where the gate of the additional MOS transistor is connected to the gates of the first and second MOS transistors and the channel constant of said additional MOS transistor is at most one-fifth the value of the channel constant associated with the remaining MOS transistors;

and the output circuit branch further comprises an additional MOS transistor connected in series between the pair of MOS transistors, where, the gate of the additional MOS transistor is connected to the source of said additional MOS transistor of said input circuit branch.

3. An MOS current mirror as defined in claims **1** or **2** where each MOS transistor is an N-channel type MOS transistor.

4. An MOS current mirror as defined in claims **1** or **2** where each MOS transistor is a P-channel type MOS transistor.

5. An MOS current mirror circuit comprising an input circuit branch and an output circuit branch wherein the input circuit branch is responsive to a reference current and the output branch mirrors the reference current to produce an output current substantially equal to said reference current, each circuit branch comprising a plurality of source-substrate connected MOS transistors wherein each MOS transistor exhibits a substantially identical threshold voltage, V_T, said input and output circuit branches interconnected such that a first interconnection will exhibit a voltage approximately equal to the quantity V_T+2V_{ON}, a second interconnection will exhibit a voltage approximately equal to the quantity V_T+V_{ON}, and a third interconnection between the input circuit branch and the output circuit branch will exhibit a voltage approximately equal to the quantity V_T+3V_{ON}.

* * * * *